

Terraform Errors & Solutions - Quick Reference Guide

Provider/Auth Issues

- No credentials / AccessDenied → Set AWS creds/env vars, fix IAM permissions.
- Unsupported provider version → Pin version in required_providers, run terraform init -upgrade.

Init/Modules

- Failed to install provider/module → Check required_providers or module source.
- Version conflicts → Pin module versions, re-init.

API/Network

- RateLimitExceeded / Throttling → Reduce parallelism (-parallelism=5), retry.
- Timeouts → Add timeouts block or increase retries.

State Management

- State lock error → Wait or terraform force-unlock.
- AccessDenied on remote state → Fix backend policy/credentials.
- State drift (console/manual changes) → Import resource or override with TF.

Resource Errors

- AlreadyExists (e.g., S3 bucket) → Use unique name/random_id.
- DependencyViolation (destroy blocked) → Destroy dependents first or target specific resource.
- Immutable field change → Resource recreated unless ignore_changes used.

Expressions & Logic

- Invalid index / missing key → Use length(), lookup(), try().
- for_each depends on unknown value → Avoid dynamic unknowns, split into multiple applies.
- Unsupported attribute → Check docs, test in terraform console.

Environments & Collaboration

- Name clashes (across dev/prod) → Use workspaces or env-specific vars.
- Local vs remote state drift → Always use remote backend (S3/GCS/Terraform Cloud).

Outputs & Secrets

- Sensitive values exposed → Use sensitive = true in output blocks.

General Debugging

- terraform fmt && terraform validate → Catch syntax issues early.
- terraform plan → Preview changes before apply.

- `TF_LOG=DEBUG` → Detailed logs for troubleshooting.
- `terraform console` → Test expressions interactively.